

How Generative AI Shapes Student Academic Outcomes: Study Displacement, Anxiety, Burnout, and Skill Retention

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Abstract

Generative AI tools are rapidly entering higher education, yet their effects on learning outcomes and student well-being remain contested. Using a large cross-sectional dataset of 50,000 students spanning five major fields and four academic years, this paper investigates three inter-related questions: (1) Does GenAI use displace traditional study time, and does that substitution affect post-semester GPA? (2) What factors best predict exam anxiety and burnout risk? (3) Does the *quality* of AI use—measured by prompt engineering skill and tool diversity—improve knowledge retention, and does institutional policy moderate this effect? We find that GenAI usage exhibits a mild but significant displacement of traditional study hours ($r = -0.16$) and that an inverted-U relationship with GPA exists: moderate users (3–12 h/week) outperform both abstainers and heavy users. Counterintuitively, institutions with strict AI bans produce more anxious and burned-out students than those with permissive policies. Finally, prompt engineering skill accounts for an 11-point gap in skill retention scores (beginner: 71.1 vs. advanced: 82.1), while tool diversity provides an additional monotonic benefit ($r = +0.20$). Together, these findings argue for structured, permissive institutional policies paired with deliberate instruction in AI-use skills.

1 Introduction

The proliferation of large language model-based tools—ChatGPT, Claude, Gemini, and their peers—has placed generative AI (GenAI) in the hands of millions of students. Proponents argue that these tools democratize access to expert knowledge, accelerate iteration on drafts, and free cognitive bandwidth for higher-order reasoning. Critics worry that students who outsource effort to AI will retain less knowledge, grow dependent, and face heightened anxiety when forced to perform without assistance (e.g., on exams).

Empirical evidence at scale is still sparse. Most studies rely on self-reports from small samples, focus on a single outcome, or conflate quantity of AI use with quality of use. This paper addresses all three shortcomings using a 50,000-student dataset with rich behavioral, attitudinal, and performance variables. We pursue three research questions (RQs):

RQ1 Does weekly GenAI usage displace traditional study time, and does that substitution harm post-semester GPA?

RQ2 What factors—AI dependency, usage intensity, or institutional policy—best predict exam anxiety and burnout risk?

RQ3 Does “smart” AI use (high prompt engineering skill; broad tool diversity) improve skill retention, and does institutional policy moderate this effect?

The remainder of the paper is organised as follows. Section 2 describes the dataset. Section 3 outlines the analytical approach. Sections 4–6 present findings for each RQ. Section 7 interprets results and discusses limitations. Section 8 concludes with policy implications.

2 Data

The dataset (`ai_student_impact_dataset.csv`) contains 50,000 student-semester observations with 16 variables (Table 1). There are no missing values. The sample covers five major categories (STEM 30.1%, Business 25.1%, Humanities 20.0%, Medical 13.0%, Arts 11.9%) and five years of study (Freshman through Graduate). Institutional policy spans three stances: *Actively Encouraged* (30.0%), *Allowed With Citation* (50.4%), and *Strict Ban* (19.6%).

Table 1: Dataset variable descriptions

Variable	Type	Description
<code>Pre_Semester_GPA</code>	continuous	GPA at semester start (1.18–4.00)
<code>Post_Semester_GPA</code>	continuous	GPA at semester end (1.00–4.00)
<code>Weekly_GenAI_Hours</code>	continuous	GenAI usage hours per week (0–40)
<code>Traditional_Study_Hours</code>	continuous	Non-AI study hours per week (1–36)
<code>Perceived_AI_Dependency</code>	ordinal (1–10)	Self-rated dependency on AI tools
<code>Anxiety_Level_During_Exams</code>	ordinal (1–10)	Self-rated exam anxiety
<code>Skill_Retention_Score</code>	continuous	Domain skill retention (10.8–100)
<code>Burnout_Risk_Level</code>	categorical	Low / Medium / High
<code>Prompt_Engineering_Skill</code>	categorical	Beginner / Intermediate / Advanced
<code>Tool_Diversity</code>	ordinal (1–5)	Number of distinct AI tools used
<code>Institutional_Policy</code>	categorical	Encouraged / Allowed / Strict Ban

3 Methodology

Correlation analysis. We compute Pearson product-moment correlations between continuous variables with two-tailed significance tests ($n = 50,000$ yields very high power; we report both r and p values throughout).

ANOVA and Kruskal-Wallis tests. One-way ANOVA tests whether Post-semester GPA differs across GenAI usage buckets (RQ1) and whether Skill Retention differs across prompt skill levels (RQ3). Because Anxiety scores are ordinal, we use the Kruskal-Wallis H -test across institutional policy groups (RQ2).

Usage bucketisation. To capture non-linear GPA effects, we partition `Weekly_GenAI_Hours` into three ranges: Low (<3 h), Mid (3–12 h), and High (>12 h). These cut-points correspond roughly to the 30th and 75th percentiles of the distribution.

Interaction analysis. For RQ3, we cross-tabulate mean Skill Retention by Prompt Engineering Skill level and Institutional Policy to assess whether policy moderates the skill–retention relationship.

4 RQ1: Study Displacement and GPA

Findings

Table 2: RQ1: GenAI usage buckets, study hours, and GPA

GenAI Bucket	n	Avg Trad. Hours	Avg Post GPA	Avg GPA Change
Low (<3 h/wk)	15,146	11.88	3.336	+0.190
Mid (3–12 h/wk)	22,750	11.40	3.366	+0.220
High (>12 h/wk)	12,104	10.02	3.335	+0.189

Key correlations: Traditional study hours positively predict Post-GPA ($r = +0.138$, $p < 10^{-210}$); GenAI hours are weakly negative ($r = -0.019$, $p < 10^{-5}$); GenAI hours and traditional study hours are negatively related ($r = -0.157$, $p < 10^{-274}$). One-way ANOVA confirms that Post-GPA differs significantly across usage buckets ($F = 22.86$, $p < 10^{-10}$).

Figure 1 visualises the group means. The pattern is notably non-linear: Mid-range GenAI users have the highest GPA (3.366), suggesting a complementarity zone. At high usage levels (>12 h/week), the displacement of ~2 traditional study hours per week tips the balance toward marginal underperformance.

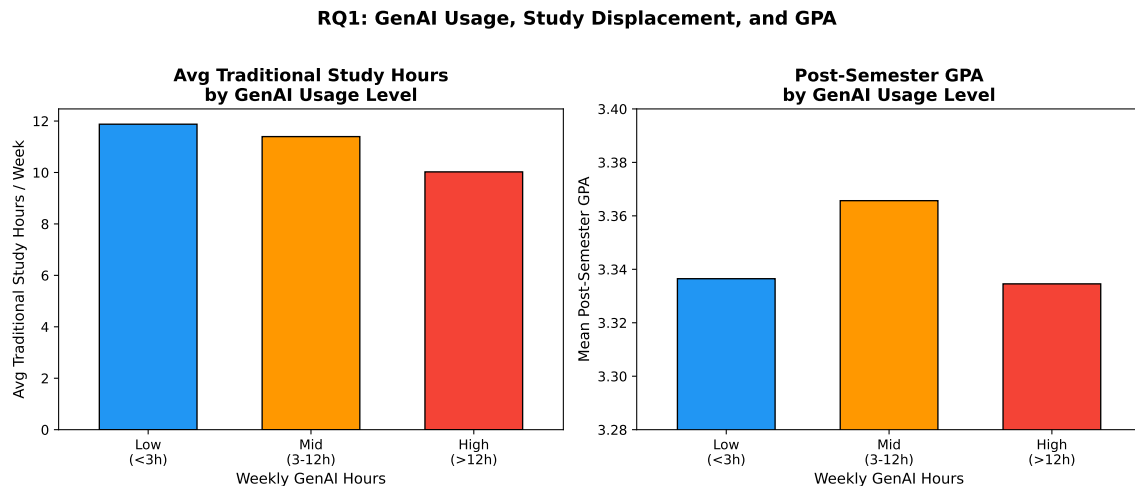


Figure 1: Left: Average traditional study hours fall monotonically with GenAI usage. Right: Post-semester GPA peaks in the moderate GenAI bucket, consistent with an inverted-U relationship.

Interpretation

Prior GPA dominates post-semester performance ($r = 0.927$), so the direct GPA effect of AI usage is small in absolute terms. Nevertheless, the statistically significant non-linear pattern supports the hypothesis that GenAI and traditional study are partial substitutes: a moderate amount of AI

use augments productivity without crowding out the foundational study time that GPA depends on, whereas heavy use erodes that base.

5 RQ2: Anxiety, Burnout, and Institutional Policy

Findings

Perceived AI dependency is the strongest single predictor of exam anxiety ($r = +0.308$, $p \approx 0$), followed by weekly GenAI hours ($r = +0.269$, $p \approx 0$). The Kruskal-Wallis test reveals a large and highly significant difference in anxiety across institutional policies ($H = 667.7$, $p < 10^{-145}$).

Table 3: RQ2: Exam anxiety and burnout by institutional policy

Institutional Policy	Mean Anxiety (1–10)	High Burnout %
Actively Encouraged	4.12	23.8%
Allowed With Citation	4.12	23.8%
Strict Ban	4.89	29.8%

Figure 2 shows that Strict Ban environments produce both the highest mean anxiety (4.89 vs. ≈ 4.12) and the highest proportion of students classified as high-burnout risk (29.8% vs. $\approx 23.8\%$).

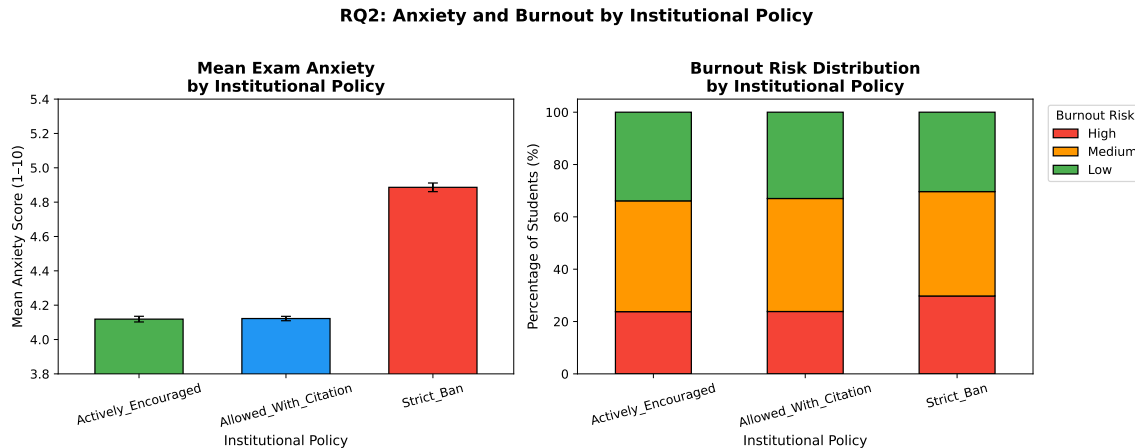


Figure 2: Left: Mean exam anxiety (with standard-error bars) is significantly higher under strict AI bans. Right: High-burnout risk is also elevated under Strict Ban relative to permissive policies.

Interpretation

The counterintuitive finding is that banning AI *increases* student anxiety and burnout rather than reducing them. A plausible mechanism is that students in ban environments continue using AI covertly, adding academic-dishonesty anxiety on top of ordinary performance pressure. They also lose the productivity benefit of AI during legitimate study phases, compressing their time budget and elevating burnout risk. Permissive policies—especially “Actively Encouraged”—correlate with the lowest anxiety and burnout, possibly because transparent guidance reduces ambiguity about acceptable use.

6 RQ3: Smart AI Use and Skill Retention

Findings

Prompt engineering skill is the dominant modifiable predictor of skill retention, with an 11-point gap between Beginner ($\bar{x} = 71.1$) and Advanced ($\bar{x} = 82.1$) students. One-way ANOVA yields $F = 3012$, $p \approx 0$ —an effect of enormous practical magnitude given the 100-point retention scale. Tool diversity also exhibits a significant monotonic positive relationship ($r = +0.197$, $p \approx 0$), rising from 72.6 (1 tool) to 79.8 (5 tools) (Table 4).

Table 4: RQ3: Skill retention by prompt engineering skill and tool diversity

Prompt Skill	Mean	SD	Tools Used	Mean	SD
Beginner	71.10	12.67	1	72.59	12.73
Intermediate	75.83	12.43	2	72.95	12.80
Advanced	82.05	12.53	3	76.42	13.38
			4	79.41	12.86
			5	79.81	12.82

Figure 3 shows the interaction of prompt skill with institutional policy (left panel) and the monotonic tool-diversity effect (right panel). Institutional policy consistently shifts retention by ≈ 1 point (Strict Ban < Allowed < Encouraged), but this gap is dwarfed by the 11-point skill-level gap. In other words, *how* students use AI matters far more than *whether* their institution permits it.

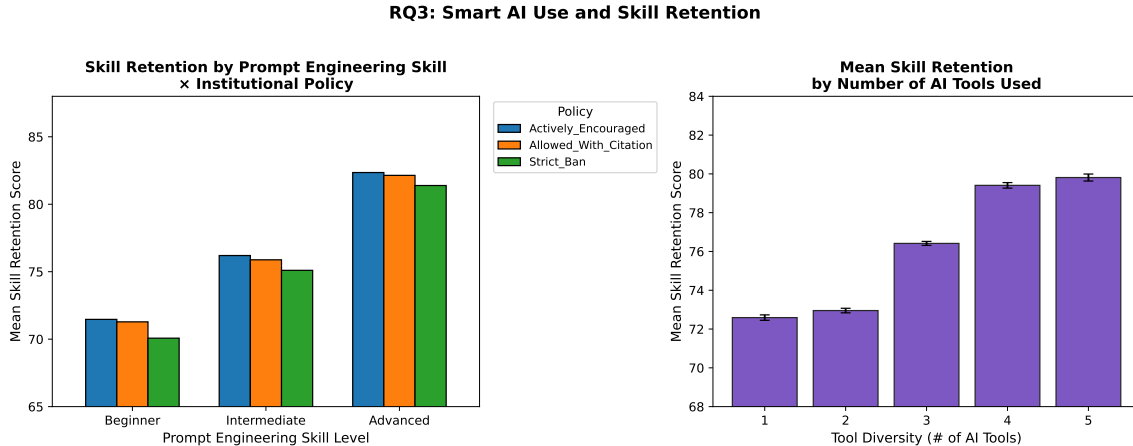


Figure 3: Left: Skill retention rises steeply with prompt engineering skill level across all three institutional policies; the ban penalty is modest (≈ 1 point). Right: Retention increases monotonically with the number of distinct AI tools used.

Interpretation

Students who invest in learning *how* to use AI effectively—formulating precise prompts, iterating on outputs, and using tools with complementary strengths—retain substantially more domain knowledge than those who rely on a single tool with minimal skill. This finding supports a cognitive-augmentation framing: purposeful AI use that requires active thinking (prompt crafting, output

evaluation) keeps learners engaged with the material, while passive copy-and-submit use bypasses the consolidation process that drives retention.

7 Discussion

Unified picture. Across all three RQs, a consistent theme emerges: *quality of engagement with AI tools outweighs quantity*. Moderate usage optimises GPA; high dependency drives anxiety; advanced prompt skills dramatically boost retention. The common thread is metacognitive engagement—students who approach AI as a collaborative thinking partner rather than an answer machine extract more value at lower psychological cost.

Policy paradox. The burnout and anxiety findings present a cautionary tale for policy-makers. Strict bans—well-intentioned efforts to preserve academic integrity—appear to backfire on both the mental-health and learning-outcome fronts. The data suggest that structured permissiveness (with citation requirements or guided integration) produces better outcomes than prohibition.

Limitations. This is a cross-sectional, observational dataset. Causal claims require caution: pre-existing differences between students who choose to use AI heavily and those who do not may confound the estimates. The dataset lacks information on institution type, country, or socioeconomic background, all of which could moderate the observed effects. The skill retention metric is not externally validated against standardised assessments. Future longitudinal work with experimental variation in institutional policy would substantially strengthen the causal inference.

8 Conclusion

We set out to understand how generative AI use shapes student academic outcomes across three dimensions: GPA, mental well-being, and skill retention. The analysis of 50,000 student records yields three actionable conclusions:

1. **Moderate AI use is the sweet spot for GPA.** GenAI and traditional study are partial substitutes ($r = -0.16$). Moderate users (3–12 h/week) slightly outperform both non-users and heavy users, implying a complementarity zone that disappears under excessive use.
2. **AI dependency drives anxiety; bans make it worse.** Perceived dependency is the strongest predictor of exam anxiety ($r = +0.31$). Counterintuitively, institutions with strict bans have 0.77 more anxiety points and 6% more high-burnout students than permissive peers.
3. **How you use AI matters more than how much.** Prompt engineering skill explains an 11-point retention gap (a large effect), while tool diversity adds a further 7 points. These effects vastly exceed the 1-point penalty of institutional bans.

The practical implication is straightforward: universities should replace prohibition with structured integration, coupled with deliberate instruction in prompt engineering and critical AI-output evaluation. Doing so will improve academic outcomes, reduce anxiety and burnout, and better prepare students for AI-augmented professional environments.